

Department, Trainer & Employee Model — Target vs. Current

LMS — QT Consultancy • Draft for review • Prepared 03/10/2026 • No code changed yet

Purpose of this document

Captures the relationship model you want, compares it with what the code does today, lists every place that would need to change, and the decisions you must make before implementation. Confirm or adjust the **Decision Points** in section 7 and tell me the scope to build.

1. Target data model (what you want)

- **Admin** — exactly one. Creates users and designates trainers of departments. Sees global compliance.
- **Employee** — can belong to **many departments** (one user many depts).
- **Trainer** — an employee promoted by the admin to **manage many departments** (one trainer many depts).
- A single department can have **many trainers** and **many employees**.
- So `user department` is a genuine **many-to-many** for every role, and the same relationship serves both trainers and employees.
- Trainer can **create courses** and **assign** them to all departments they manage, reaching all employees in those departments.
- Employee **catalog** shows every course assigned directly to them plus every course assigned to **any** department they belong to.
- Removing a person from a department revokes that department's visibility for them, but **must never delete courses they created**.

2. Roles & capabilities

Role	Scope	Key capabilities
Admin	Company-wide (all departments)	Create/edit users; promote employee trainer; assign trainers to departments; assign courses anywhere; view full compliance (who is in which dept, which courses to whom).
Trainer	Only the departments they manage	Create courses; assign to any of their managed departments and their employees; view compliance/stats for their departments only.
Employee	Belongs to one or more departments	See catalog of courses assigned to self or to any department they are in; complete lessons/quizzes; earn certificates.

3. Current implementation (what the code does today)

3.1 Database tables (relevant)

Table	Department linkage	Cardinality today
users	Single column <code>department_id</code> FK departments (nullable, ON DELETE SET NULL)	One user at most one department

trainer_departments	(trainer_id, department_id), UNIQUE pair	Trainer many depts; dept many trainers (already many-to-many)
departments	id, name	—
course_assignments	(course_id, user_id) rows created per target user	Course many users

3.2 How each flow behaves now

- **Employee membership:** fixed by `users.department_id` — a single *home* department only. Multiple departments are **not supported**.
- **Trainer membership:** managed via `trainer_departments`; a trainer's own home dept is also auto-included (recent fix).
- **Users tab (admin.js / AdminController):** the Department column reads the *home* dept; a dropdown writes `users.department_id` (single value).
- **Departments tab (+Trainer):** writes `trainer_departments` — a **different** table, so it does not appear in the Users tab. This is the mismatch you noticed.
- **Course targeting by dept (AssignmentController):** selects employees `WHERE department_id = ? AND role='employee'` — home dept only.
- **Trainer authorization:** allowed to target a dept if a `trainer_departments` row exists OR it is their home dept.
- **Auth / JWT / login / certificates:** return a **single** `department_id` / `department_name`.
- **Admin compliance:** per-department employee count and stats computed from `users.department_id` (home dept) joined to `course_assignments`.

4. Gap analysis — target vs. current

Capability	Status	Why / what is missing
Dept has many trainers	Supported	<code>trainer_departments</code> already many-to-many.
Trainer manages many depts	Supported	<code>trainer_departments</code> + home-dept union.
Employee in many depts	Not supported	Only one <code>users.department_id</code> . Needs a general <code>user_departments</code> join table.
Catalog shows courses from any of the employee's depts	Partial	Works only for the single home dept; fails once multi-membership exists.
Assign-to-dept reaches all members	Partial	Targets home-dept employees only; must target join-table members.
Admin compliance lists who (trainer/employee) is in which depts, and course people	Partial	Only reflects home dept; trainer managed list lives in a separate table.
One unified dept source of truth (Users tab == Departments tab)	Not supported	Two tables (<code>users.department_id</code> and <code>trainer_departments</code>) drift apart.
Remove-from-dept revokes access but keeps created courses	Safe	Removal only edits membership; it does not delete courses. Must preserve this in rewrite.

5. Code that must change if we adopt many-to-many

Backend lines require a Java restart to activate; statics hot-sync.

Area	File : line	Change required
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Schema	new V5... migration	Create <code>user_departments(user_id, department_id, ...)</code> ; backfill from <code>users.department_id</code> and <code>trainer_departments</code> .
Create user	AdminController:88	Write one row in <code>users</code> (no single dept) + rows in <code>user_departments</code> .
Update role /dept	AdminController:96–121	Replace single <code>department_id</code> write with set-add/remove of membership rows.
Roster read	AdminController:49	Return each user's list of departments (name + id) instead of one.
Compliance	AdminController:34	Count members via join table, split by role; decide double-count (see D3).
Department list / members	DepartmentController: 5,11	Read members and per-dept counts from join table.
Add/remove member	DepartmentController: 18,21	Insert / delete a <code>user_departments</code> row (must not null a user's other depts).
Assign /unassign trainer	DepartmentController: 22,23	Merge into same table (or view) used by membership.
Trainer "my depts"	DepartmentController:6	Read all managed depts from join table; drop home-dept union hack.
Course targeting	AssignmentController:47	Target <code>user_id</code> IN (SELECT ... FROM <code>user_departments</code> WHERE <code>department_id=?</code>).
Trainer auth on dept	AssignmentController:49	Single join-table lookup for "can this trainer assign to dept X".
Login / session	AuthController:18,34 • JwtAuthenticationFilter:24	Return list of depts (or a chosen primary). Affects every page that shows the user's dept.
Certificate	ProgressController:7,9	Pick which dept prints when the user is in several (see D2).
Users tab UI	admin.js:423–425,454,181–182	Single dropdown multi-select; filter matches "belongs to dept X".
Departments tab UI	admin.js:291,350	Read members/trainers from unified source.
Other dept displays	navbar / dashboard / profile	Handle a list (e.g. primary + "+N more").

6. Removal & cascade rules (to preserve)

- Removing a user from **Dept X** deletes only that membership row; their other departments and their `course_assignments` /progress stay.
- A trainer losing Dept X: X's courses/employees no longer appear in their trainer views; the **courses they authored remain** and stay assigned to whoever already received them.
- Deleting a **department** must not delete courses; assignments already made should be preserved or explicitly handled (see D5).
- Demoting a trainer to employee should drop their *management* rights but can keep or remove membership depending on D4.

7. Decision points (please answer before I build)

D1. Home/primary department. Keep `users.department_id` as one "primary" dept (used for login/JWT /certificate default display) while `user_departments` holds all memberships — OR fully retire `department_id` ? (Keep = lower risk, side-by-side. Retire = cleaner but touches auth/JWT/certificates broadly.)

D2. Certificate dept. When a user belongs to several departments, the certificate shows: primary dept / most recent / a chosen one / comma list?

D3. Compliance double-count. A user in two departments: counted in **both** departments' employee totals (per-dept view) but **once** in the company-wide total? This changes the compliance percentage math.

D4. Trainer demotion. When trainer employee, remove their managed-department memberships, or keep them as a plain employee member of those depts?

D5. Department deletion. Deleting a department: cascade remove memberships only, or also revoke the assignments created through it?

D6. Scope for first build. Full many-to-many now, or phase 1 = add `user_departments` for employees + unify the Users/Departments tabs, and phase 2 = rewrite compliance/targeting/certificate?

End of document. Nothing has been implemented from this yet; awaiting your decisions in section 7 and the scope you want built.